

Mahmoud Hussain

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Education

United International University

Bachelor of Science in Computer Science and Engineering

Dhaka, Bangladesh

Expected 2027

Bangladesh International School English Section

*Cambridge International Curriculum — A-Level (2021),
IGCSE / O-Level (2018)*

Jeddah, Saudi Arabia

Technical Skills

Programming: Python, C++, Java, JavaScript, TypeScript, C

AI & Machine Learning: PyTorch, Federated Learning, MobileNetV3, Grad-CAM, Flower

Application Development: FastAPI, React Native, Expo, OpenAPI

Tools & Monitoring: Git, GitHub, Linux, CodeCarbon, psutil

Languages: Bengali (Fluent), English (Fluent), Urdu (Conversational), Arabic (Basic comprehension)

Selected Technical Projects

GreenScan — AI Waste Classification & Federated

Learning Platform

Python, PyTorch, MobileNetV3, Flower, FastAPI, Grad-CAM, CodeCarbon, psutil

- Developed a waste-classification platform using PyTorch and MobileNetV3, supporting centralized training workflows and distributed federated learning via Flower.
- Integrated Grad-CAM for model explainability, FastAPI for inference endpoints, and CodeCarbon with psutil for monitoring compute utilization and estimated CO₂ emissions.

Multimodal Human Sensing — Mobile Sensing & AI

Framework

TypeScript, React Native, Expo

- Built an Android background-sensing application with React Native and Expo to collect real-time tri-axial accelerometer streams, audio levels, and network connectivity states.
- Implemented bounded FIFO sensor ring-buffers and sampling-rate telemetry, transiently processing audio metrics in memory without persistent disk storage or network transmission.

API Sentinel — API Security & Monitoring Platform

Python, OpenAPI

- Architected a modular Python-based API security testing and monitoring platform featuring diagnostic reporting, automated test execution, and dashboard components.
- Structured functionality into reusable packages, integrating automated test suites and standardized OpenAPI-compliant documentation.

Research Experience

X-FIBF: Explainable File Integrity Behavioral

Fingerprinting for Ransomware Detection

Cybersecurity, Explainable AI, Temporal Deep Learning

- Contributed to research combining static PE analysis, file-integrity behavioral fingerprinting, and temporal API-call learning for ransomware detection.
- Worked with an MHSA-BiLSTM approach and attention-based explainability mechanisms for analyzing dynamic ransomware behavior.

GreenFed-WasteNet: Carbon-Aware Federated

Baselines for Waste Classification

Federated Learning, Green AI, Computer Vision, Sustainable Computing

- Contributed to research addressing class imbalance, open-set uncertainty, inference efficiency, and estimated environmental cost in federated waste classification.
- Worked with an eight-class dataset of 65,669 images and evaluated EfficientNet-B0 and MobileNetV3 baselines.

Professional Experience

Luminoso Events

2D Graphics Designer

Bangladesh

2022–Present

- Create visual concepts and production-ready 2D graphics for wedding and event projects, adapting assets to client specifications and physical installation constraints.

Competitions

FutureMakers — Grameenphone

Participant / Team Member

- Selected for Round 2 of Grameenphone's nationwide AI-focused innovation competition from an initial field of 782 participating teams.